ENTERPRISE WI-FI FLEET & CAPTIVE BILLING SYSTEM

WavePass Operations & Systems Architecture Manual

A Complete Operational Field Guide & Technical Blueprint Covering MikroTik RouterOS v7 Hardware Engineering, Android Mobile POS Operations, Multi-Tenant Captive Web Portals, and NestJS Cloud Server Administration.

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September 2026	**NestJS + BullMQ + Redis**	**Android POS + Next.js 14**	**Paystack Dedicated Accounts**

CORE ARCHITECTURAL PILLARS

1. MikroTik RouterOS v7 Edge Local hardware execution via REST API. Zero cloud dependency for local hotspot authentication, dynamic bandwidth queues, and walled-garden pass-through.	**2. Mobile POS & Thermal Printing** Android tablet/phone application. Direct Wi-Fi auto-discovery (192.168.88.1), 1-tap pass generation, synchronous hardware push, and ESC/POS Bluetooth receipt cutouts.
3. Next.js 14 Multi-Tenant Portal Wildcard subdomain routing (venue.nexawavepass.com). Instant MAC authentication, digital Paystack card/USSD checkout, and camera voucher QR redemption.	**4. NestJS Cloud & Async Queue** PostgreSQL persistence via Prisma ORM. BullMQ worker engine over Upstash Serverless Redis for resilient retry-backoff, fleet auditing, and DVA cashouts.

OPERATIONAL BEST PRACTICE: **Executive Summary:** WavePass is a commercial-grade telecommunications and billing ecosystem designed to solve the historical reliability, operational fragmentation, and manual labor bottlenecks associated with MikroTik Hotspot networks in retail, hospitality, enterprise, and public venues. By decoupling local high-speed hardware authentication from cloud billing and fleet analytics, WavePass enables sub-second cash pass sales, frictionless self-service digital payments via Paystack, automated multi-venue reconciliation, and resilient offline operation during upstream ISP failure.

Chapter 1: System Overview & Architecture Philosophy

1.1 The WavePass Problem Statement: Modernizing Beyond Mikhmon

For over a decade, network operators across emerging markets and hospitality sectors relied on tools like **Mikhmon** and legacy FreeRADIUS servers to manage MikroTik RouterOS hotspots. While functional for basic local voucher generation, legacy tooling suffered from fundamental architectural flaws:

- **Local-Only Confinement:** Mikhmon runs as a local PHP script that must sit on the same subnet as the router or require risky port forwarding (Winbox port 8728/8729) exposed to the public internet.
- **No Digital Self-Checkout:** Customers were entirely reliant on physical cashiers selling paper vouchers. There was no native mechanism for guests to pay via debit cards, bank transfers, or USSD and unlock Wi-Fi automatically.
- **Fragile Synchronization:** Cloud RADIUS solutions introduced unacceptable latency: if the venue's internet uplink flapped, walk-up customers could not buy passes or authenticate because the cloud was unreachable.
- **Zero Multi-Venue Telemetry:** Operators managing multiple cafes, hotels, or campuses were forced to maintain separate, disconnected databases with zero consolidated revenue tracking or fleet health auditing.

WavePass resolves these bottlenecks through a Hybrid Edge-Cloud Topology. Time-sensitive user authentication, dynamic rate limiting, and walk-up cash pass provisioning occur directly on the physical MikroTik router hardware via its modern RouterOS v7 REST API. Concurrently, billing reconciliation, multi-tenant fleet administration, and Paystack digital payments are coordinated by a resilient cloud backend.

1.2 The Hybrid Edge-Cloud Model: Local LAN vs Remote Tunnel

To deliver both sub-millisecond local performance and centralized cloud control, WavePass implements two distinct communication pathways between clients and hardware:

FEATURE / ATTRIBUTE	LOCAL SUBNET DIRECT (LAN)	REMOTE CLOUD / WIREGUARD TUNNEL
Target Endpoint	`http://192.168.88.1:80` (or local IP)	`http://10.8.0.2:80` or WireGuard tunnel
Round-Trip Latency	< 2 ms (Instantaneous Wi-Fi / Ethernet)	25 – 80 ms (Dependent on WAN transit)
Offline Resiliency	**100% Operational** during uplink failure	Requires active upstream ISP internet connection
Primary Use Case	Cashier POS sales, batch printing, guest portal	Remote admin monitoring, webhooks, cloud sync
Security Model	Restricted to authenticated venue Wi-Fi clients	End-to-end encrypted private tunnel peering

> **CRITICAL REQUIREMENT:** WavePass utilizes the **Dual Connection Engine** in the mobile app. When an operator is on the shop's Wi-Fi, the app automatically executes REST commands directly against 192.168.88.1. When the operator is off-site, the app seamlessly routes telemetry through the secure remote tunnel.

1.3 System Component Matrix

TIER / SUBSYSTEM	TECHNOLOGY STACK	NETWORK LAYER	FUNCTIONAL RESPONSIBILITY
Edge Router Hardware	MikroTik RouterOS v7.12+	Venue Gateway (LAN)	Captive redirection, MAC auth, bandwidth throttling, hotspot queues.
Mobile POS App	Flutter 3.x (Android)	Local Wi-Fi / WAN	Cash pass sales, router auto-discovery, thermal printing, device kicking.
Captive Web Portal	Next.js 14, Tailwind CSS	Public Cloud (Vercel)	Multi-tenant captive landing, Paystack checkout, voucher QR scan.
Cloud Backend API	NestJS, TypeScript, Prisma	Cloud Run / VPS	Core REST API, payment webhook verification, session tracking, DVA billing.
Async Task Queue	BullMQ, Upstash Redis	Cloud Serverless TLS	Decoupled hardware provisioning, exponential retry backoff, cleanup.
Payment Settlement	Paystack API (DVA / Card)	PCI-DSS Public API	Dedicated virtual accounts, debit cards, USSD, automated payout transfers.

1.4 End-to-End Transaction & Packet Lifecycle

Understanding how data traverses between the guest device, physical router, and cloud services ensures rapid troubleshooting:

STAGE	GUEST / CLIENT ACTION	EDGE ROUTER HARDWARE ACTION	WAVEPASS CLOUD / POS ACTION
1. Association	Guest connects to Wi-Fi SSID.	DHCP assigns 192.168.88.X; marks unauthenticated.	Idle.
2. Interception	Device opens browser (or CNA triggers).	Captive engine 302 redirects to portal with MAC param.	Next.js middleware injects venue branding headers.
3. Purchase	Pays via Paystack or enters cash voucher.	Walled garden permits traffic to Paystack/API.	Paystack confirms transaction via webhook.
4. Activation	Receives 'Internet Unlocked' feedback.	User created in /ip hotspot user; queue attached.	BullMQ queue marks session ACTIVE in PostgreSQL.
5. Expiration	Duration expires (e.g. 1 hour).	RouterOS disconnects client; redirects to captive.	Cron marks session EXPIRED; prompts re-up checkout.

Chapter 2: MikroTik RouterOS Hardware Engineering & Setup

2.1 Supported Hardware Specifications

WavePass is engineered exclusively for **MikroTik RouterOS v7.12 or higher**. RouterOS v7 introduced a native, high-performance REST API running over HTTP/HTTPS that replaces the legacy proprietary binary API (port 8728). Recommended hardware tiers based on concurrent client capacity:

VENUE SIZE	RECOMMENDED MODEL	CPU / RAM SPECIFICATIONS	CONCURRENT USERS
Small Venue (Cafe, Salon, POS)	MikroTik hEX (RB750Gr3)	Dual-Core 880 MHz (MMIPS), 256 MB RAM	**Up to 75 active devices**
Medium Venue (Restaurant, Hotel)	hEX S / hAP ax2 / ax3	Quad-Core 864 MHz (ARM), 1 GB RAM	**75 – 250 active devices**
Large Venue (Campus, Stadium)	RB5009UG+S+IN / CCR2004	Quad-Core 1.4 GHz (ARM64), 1 GB – 4 GB RAM	**500 – 2,500 active devices**

2.2 Subnet Architecture & Network Plan

By default, factory MikroTik routers bind Ether2 through Ether5 into a master LAN bridge (`bridge1`). WavePass establishes the following standardized network topology:

- **WAN Uplink (Ether1):** Configured as DHCP Client receiving upstream public IP/gateway from the ISP, or PPPoE client.
- **LAN Hotspot Gateway (bridge1):** Fixed IP `192.168.88.1/24`. All customer devices connect to this bridge.
- **DHCP Hotspot Pool:** Range `192.168.88.10 - 192.168.88.254` with a 30-minute lease time to rapidly recycle IPs from transient foot traffic.
- **DNS Service:** Router acts as primary caching resolver (`192.168.88.1`) redirecting upstream to Google/Cloudflare (8.8.8.8, 1.1.1.1).

2.3 Enabling the REST API & Admin Credentials Standard

To enable WavePass mobile and cloud software to communicate with the router, the native **www** (HTTP) or **www-ssl** (HTTPS) service must be active on the device. Execute the following command in Winbox Terminal:

```
/ip service set www disabled=no port=80
/ip service set www-ssl disabled=no port=443
```

> **OPERATIONAL BEST PRACTICE:** **Default Admin Credentials Standard:** WavePass standardized on the default MikroTik **admin** account (with blank password out-of-box, or the operator's customized admin password). WavePass deliberately does NOT inject extra 'wavepass' users into RouterOS. This eliminates password sync mismatches, preserves Winbox administrative clarity, and prevents router lockout.

2.4 The Automated Single-Command Setup Script

WavePass provides a comprehensive auto-configuration script (`docs/wavepass-setup.rsc`). When pasted into Winbox Terminal, it completes the entire hotspot configuration in under 30 seconds:

```
# 1. Ensure Web REST API is active on Port 80
/ip service set www disabled=no port=80

# 2. Configure WavePass Hotspot Profile
/ip hotspot profile add name=wavepass html-directory=flash/hotspot login-by=http-chap,http-pap,mac-cookie

# 3. Create Dynamic Rate-Limiting Plan Profiles
/ip hotspot user profile add name=profile_1h rate-limit="2M/1M" session-timeout=1h shared-users=1
/ip hotspot user profile add name=profile_12h rate-limit="3M/1M" session-timeout=12h shared-users=1
/ip hotspot user profile add name=profile_1d rate-limit="4M/2M" session-timeout=1d shared-users=1
/ip hotspot user profile add name=profile_7d rate-limit="5M/2M" session-timeout=7d shared-users=1
/ip hotspot user profile add name=profile_30d rate-limit="5M/2M" session-timeout=30d shared-users=1

# 4. Walled Garden Rules (Allow Payment & Captive Portal Without Authentication)
/ip hotspot walled-garden ip add dst-host=*paystack.co action=accept
/ip hotspot walled-garden ip add dst-host=*paystack.com action=accept
/ip hotspot walled-garden ip add dst-host=*nexawavepass.com action=accept
/ip hotspot walled-garden ip add dst-host=*wavepass.com action=accept
/ip hotspot walled-garden ip add dst-port=53 protocol=udp action=accept
```

2.5 Critical Hardware Nuance: FastTrack vs Hotspot Queues

CRITICAL MIKROTIK GOTCHA: Default RouterOS configurations enable **FastTrack** firewall rules (`action=fasttrack-connection`) to maximize throughput on low-power CPUs. However, FastTrack bypasses the Linux queuing discipline entirely! If FastTrack is active on hotspot packets, **user rate limits (e.g. 2M/1M) will be completely ignored** and users will consume full bandwidth. The WavePass setup script automatically resolves this by placing a dummy accept filter rule ahead of FastTrack for the hotspot subnet, ensuring dynamic Simple Queues are strictly enforced.

Chapter 3: Operator Mobile POS Application Manual (Android)

3.1 Initial Setup & Operator Authentication

The WavePass Android mobile app (`WavePass-Android`) is the operator's central point of sale, hardware telemetry terminal, and cashier tool. Built with Flutter, it operates natively on Android tablets, POS handhelds (e.g. Sunmi, Telpo), and standard smartphones.

1. Launch the app and sign in with your WavePass operator email and password.
2. The app communicates with Supabase Auth and fetches the operator's primary venue profile, including name, logo, pricing plans, and registered router fleet.
3. All transactions and passes created in the session are automatically scoped to the active venue ID.

3.2 Router Onboarding & Auto-Discovery

Adding and pairing a new MikroTik router can be achieved in under 60 seconds through two primary workflows:

ONBOARDING METHOD	STEP-BY-STEP OPERATOR PROCEDURE	TECHNICAL EXECUTION DETAILS
Auto-Find on Wi-Fi (Recommended)	1. Connect phone Wi-Fi to the router network. 2. Open **Router Setup** (`/setup-router`). 3. Tap **'Auto-Discover Router'**. 4. When detected, tap **'Install Hotspot'**.	The app sends concurrent HTTP probes to `192.168.88.1` and `192.168.1.1`. Upon receiving 200 OK JSON with RouterOS fields, it displays the hardware model, uptime, and latency, then pushes hotspot config via REST API.
Barcode Scanner (Box Serial Setup)	1. Tap **'Scan Router'** from Dashboard. 2. Scan the barcode on the router packaging. 3. Review detected Serial Number. 4. Tap **'Auto-Configure'** or **'Copy Script'**.	Registers router entity with default local gateway (`http://192.168.88.1`) and local mode. Prevents non-existent tunnel errors and guides operator to finalize setup over Wi-Fi.

3.3 Cash Pass POS Terminal (`SellPassScreen`)

For counter staff selling vouchers to walk-in customers, the **Sell Pass** screen provides a frictionless, high-speed checkout experience:

- **Select Pricing Tier:** Tap from venue-configured plans (e.g., 1 Hour = ■300, 12 Hours = ■1,000, 24 Hours = ■1,500).
- **Instant Synchronous Hardware Push:** When the operator taps **'Confirm & Sell'**, the app does NOT wait for cloud queue processing. Using `provisionVoucherDualRoute()`, it creates the hotspot user account directly on the physical router hardware in under **10 milliseconds**.
- **Hardware Status Badge:** Displays visual confirmation on screen: **'Live on Router Hardware (LAN Direct)'** or **'Live on Router Hardware (Cloud Tunnel)'**.

3.4 Bluetooth ESC/POS Thermal Printing & Voucher Cutouts

Once a pass is sold, the app triggers thermal receipt printing via Bluetooth (58mm or 80mm roll). The printed voucher cutout includes:

1. Venue Name and Logo at the top.
2. Wi-Fi Network Name (SSID) instructions.
3. High-contrast QR Code: scanning automatically opens the captive portal with code prefilled.
4. Clean Human-Readable Voucher Code (e.g. `WP-9X2A-K74D`, formatted with no ambiguous O/0/I/1 characters).
5. Plan validity duration, price, and expiration date.

3.5 Batch Voucher Production (`BatchVouchersScreen`)

For venues that sell pre-printed vouchers at retail counters, front desks, or gift shops, the Batch screen enables generating **10 to 500 vouchers** simultaneously. The batch engine compiles codes, pushes every user account onto the physical router hardware synchronously, records them in the cloud database, and exports a clean multi-column PDF sheet ready for standard desktop printer cutting.

3.6 Active Hosts Monitoring & Device Kick

In the **Active Devices** view (`/active-devices`), operators have real-time visibility into all currently connected devices. The app queries `/rest/ip/hotspot/active` to display device MAC address, assigned IP, active uptime, bytes downloaded, and bytes uploaded. If a rogue device is detected, tapping **'Kick Device'** immediately calls HTTP DELETE on the router to terminate the connection.

3.7 Operator Wallet, Reconciliation & Payouts

The **Wallet** view (`/wallet`) tracks cumulative sales split into **Cash Sales** (collected directly at the counter) and **Digital Online Sales** (collected via Paystack into the venue's Dedicated Virtual Account). When withdrawing earnings, operators request a payout to their registered NUBAN account. Payouts require two-factor admin password confirmation before being dispatched via Paystack Transfers.

Chapter 4: Guest Captive Web Portal Playbook (Web)

4.1 Multi-Tenant Subdomain Architecture (`middleware.ts`)

The guest-facing portal (`WavePass-Web`) is powered by a Next.js 14 serverless architecture. Every venue in the ecosystem is assigned a unique branded subdomain, e.g. `lagos-bistro.nexawavepass.com`. The Next.js edge middleware inspects incoming requests and dynamically injects venue context:

1. Middleware extracts the subdomain slug (`lagos-bistro`) from the `Host` header.
2. It queries the backend cache (`/api/v1/venues/by-subdomain/:slug`) to retrieve the venue's name, brand logo, and active pricing plans.
3. When guests visit the root domain (`/`), they are seamlessly redirected to `/portal` with venue headers applied.

4.2 The Guest Captive Experience: Connecting to Wi-Fi

When a customer selects the venue Wi-Fi on their phone or laptop, the operating system's Captive Network Assistant (CNA) triggers automatically (e.g. Apple CNA on iPhone/Mac, Android captive browser, Windows network login prompt):

- MikroTik Hotspot intercepts initial unauthenticated HTTP traffic and redirects the browser to the portal with captive parameters appended: `http://venue.nexawavepass.com/portal?mac=AA:BB:CC:DD:EE:FF&ip=192.168.88.245&link-login=http://192.168.88.1/login`.
- The portal auto-detects the client's device MAC address without requiring the user to look it up in system settings.
- The guest is presented with two seamless activation options: **Digital Online Payment** or **Voucher Code Redemption**.

4.3 Digital Self-Checkout via Paystack

Guests who do not have cash can purchase internet directly on their phone using Paystack:

1. The guest taps their desired plan tile (e.g., **'3 Hours — ■500'**).
2. The guest enters an optional email address for their payment receipt.
3. Tapping **'Pay with Paystack'** initializes an order and opens the secure Paystack checkout modal supporting Debit/Credit Cards, Bank USSD codes, and instant Bank Transfer via Dedicated Virtual Accounts (DVA).
4. Once Paystack confirms payment, a webhook fires to the WavePass backend. The backend immediately pushes the guest's MAC address to the router hardware, unlocking high-speed internet without the user ever typing a password.

4.4 Cash Voucher QR Scans & Network Bypass Mechanics

When guests buy a cash voucher from the counter, they can either manually type the 12-character code (`WP-XXXX-XXXX`) or simply point their phone camera at the printed QR code on their receipt. The QR code encodes: `http://venue.nexawavepass.com/login?code=WP-XXXX-XXXX`.

> **OPERATIONAL BEST PRACTICE:** **Voucher QR Scan Bypass:** Historically, mobile cameras scanning QR codes over cellular or unauthenticated Wi-Fi lacked captive MAC parameters, causing middleware to bounce them to `/network-restricted`. WavePass engineered a dedicated bypass: whenever a request carries a `code` or `voucher` parameter, the middleware permits direct access to `/portal`, allowing instant one-tap redemption.

4.5 Direct Hotspot Login Link (`http://192.168.88.1/login`)

To guarantee 100% connectivity even if cloud webhook latency occurs, the portal generates a direct link upon voucher redemption: `http://192.168.88.1/login?username=CODE&password=CODE`. If the customer is on venue Wi-Fi, tapping this link authenticates the device directly against the MikroTik router's internal HTTP login server with zero cloud delay.

4.6 Real-Time Session Telemetry (`/success` & Portal Ticker)

Once authenticated, the guest is directed to the **Success & Session Status** screen. A live, client-side polling ticker displays remaining time countdown (`HH:MM:SS`), data quota consumed vs remaining (e.g. `1.2 GB / 5.0 GB`), and current download speed. When the session is within 10 minutes of expiry, an alert banner gently prompts the user to renew.

Chapter 5: Cloud Server Operations & Backend Administration

5.1 NestJS Modular Architecture & Tech Stack

The WavePass backend (`WavePass-Backend`) is built on **NestJS**, enterprise TypeScript, and **Prisma ORM** connected to PostgreSQL. The modular layout enforces clean separation of concerns:

- **RoutersModule:** Hardware entity lifecycle, health probing (`getHealth`), reboot triggers, and dynamic provision script generation.
- **MikrotikModule:** Thin HTTP adapter (`MikrotikAdapter`) interfacing RouterOS v7 REST endpoints over private LAN or WireGuard tunnels.
- **VouchersModule:** Cryptographic human-readable voucher generation, optimistic concurrency row-locking for single-use redemption, and batch compilation.
- **PaymentsModule:** Paystack API client, Dedicated Virtual Account (DVA) management, and webhook HMAC signature verification.
- **CashoutsModule:** Two-factor password-authorized payout pipeline transferring venue earnings to verified bank NUBAN accounts.

5.2 Relational Data Models & Prisma Schema Architecture

ENTITY MODEL	PRIMARY KEY / INDEXES	CORE SCHEMA ATTRIBUTES	RELATIONAL MAPPING
Venue	`id` (UUID), `slug` (Unique)	name, logoUrl, primaryContact, address, timezone	1:N Routers, Plans, Vouchers, Orders, Cashouts
Router	`id` (UUID), `venueId`	name, endpoint, connectionMode, status, lastSeen	N:1 Venue, 1:N Sessions
Plan	`id` (UUID), `venueId`	name, priceMinor, durationSeconds, dataLimitBytes, rateLimit	N:1 Venue, 1:N Orders, Vouchers
Voucher	`id` (UUID), `codeHash` (Unique)	displayCodeEnc, status (ISSUED/REDEEMED/EXPIRED), redeemedMac	N:1 Venue, Plan, Optional Payment, 1:N Sessions
Session	`id` (UUID), `routerId, mac`	startedAt, expiresAt, status (ACTIVE/CLOSED), bytesTotal	N:1 Router, Venue, Optional Voucher/Order

5.3 BullMQ & Upstash Serverless Redis Queue Engine

To prevent slow network responses from blocking the public API during high-volume periods, all cloud-to-router hardware provisioning is queued asynchronously via **BullMQ** backed by **Upstash Serverless Redis** over TLS (`REDIS_URL`):

- **Decoupled Execution:** When Paystack webhooks fire, the API immediately confirms receipt (200 OK) to Paystack and enqueues a provisioning job.
- **Exponential Backoff & Retries:** If the router is temporarily unreachable (e.g. Wi-Fi channel flap), BullMQ retries up to 5 times with exponential backoff (1s, 2s, 4s, 8s, 16s).
- **Dead Letter Inspection:** Failed jobs are preserved in the failed queue for admin inspection without crashing the application.

5.4 Paystack Single-Key DVA Architecture & Webhook Security

WavePass uses a centralized **Single-Key Platform Model**: the platform owns one master `PAYSTACK_SECRET_KEY`, relieving individual merchant venues from having to register their own Paystack developer accounts. Incoming webhooks are verified via constant-time HMAC-SHA512 comparison against the raw request buffer to prevent spoofing.

5.5 Production Deployment with Docker Compose & Systemd

```
# Build and start production stack with Docker Compose
docker-compose up -d --build

# Verify container health and logs
docker-compose ps
docker-compose logs -f api
```

Chapter 6: Troubleshooting Runbook & Diagnostics Matrix

6.1 Field Failure Diagnostic Matrix

Quick reference table for resolving the most frequent field anomalies encountered by venue technicians and operators:

SYMPTOM / ERROR	PROBABLE ROOT CAUSE	DIAGNOSTIC TEST	IMMEDIATE RESOLUTION
Router Shows ONLINE but Refuses Connect	Endpoint set to Vercel wildcard domain instead of local gateway; HTML 200 returned.	Check endpoint in app. Probing returned HTML redirect.	Change endpoint to `http://192.168.88.1` with connectionMode 'local'. Verified in PR #10 & #21.
Guest Bounced to Network Restricted	Mobile camera QR scan lacked captive MAC query parameter.	Check URL in browser. Does it contain `?code=WP-...`?	Update web middleware to bypass restrictions on `hasVoucherParam`. Verified in PR #6.
Voucher Sold But Device Cannot Connect	Hotspot user account was not created on hardware or username was randomized.	Run in Winbox: `/ip hotspot user print`.	Ensure username equals the voucher code. Sell Pass screen pushes directly via LAN REST.
Rate Limits Ignored (Max Speed)	RouterOS FastTrack firewall filter rule active on hotspot bridge.	Check `/ip firewall filter print` for fasttrack action.	Insert dummy accept rule ahead of FastTrack rule in filter chain.
Thermal Printer Disconnects	Bluetooth connection timeout or paired device address mismatch.	Check Printer Settings in app. Bluetooth status indicator.	Re-pair ESC/POS printer in Android Settings and reconnect in app.
Paystack Webhook Dropped	Upstream ISP connection failure or port 443 firewall block.	Inspect BullMQ queue and NestJS logs for webhook HMAC receipt.	Trigger admin manual reconciliation (`POST /api/v1/admin/reconcile`).

6.2 Deep Dive: Eliminating the Barcode Scan False 'ONLINE' Trap

The Anomaly: When operators scanned a router box serial number, the app registered `https://tunnel.nexawavepass.com/$cleanSerial`. Because `*.nexawavepass.com` is a wildcard DNS pointing to Vercel, Node's `fetch()` followed a 307 redirect and received HTTP 200 OK with an HTML document. The backend adapter checked `res.ok`, which evaluated to `true`, falsely declaring the router 'ONLINE' even though no hardware was connected!

The Resolution (PR #10 & PR #21):

1. `MikrotikAdapter.testConnection()` now strictly checks `content-type: application/json` and validates RouterOS hardware fields (`platform, board-name, version`). HTML redirects return `false`.
2. The Android Barcode Scanner defaults new routers to local gateway `http://192.168.88.1` and directs operators to connect to Wi-Fi to Auto-Configure in 1 tap.

6.3 Emergency Offline Cash Mode

If the venue's upstream ISP experiences an outage, **the venue does not lose money**. Operators can continue selling cash passes via the Android app. The app connects to `http://192.168.88.1` over the local Wi-Fi without needing an internet connection, provisioning passes directly onto the router. When the ISP connection recovers, the cloud worker automatically reconciles and syncs session logs.

> **OPERATIONAL BEST PRACTICE: Weekly Maintenance Checklist:**
> 1. In Winbox, check CPU Load and Free Memory under System > Resources.
> 2. Review expired hotspot users (`/ip hotspot user remove [find comment="wavepass-expired"]`).
> 3. In the mobile app, review active cashout balances and verify thermal printer paper roll levels.